

Indian Statistical Institute

Mid semester examination

Statistics 1: Data Exploration
9 October 2025, 11am – 1pm

Bachelor of Statistical Data Science
Total points: 30

INSTRUCTIONS

Write your name and roll number in CAPITAL LETTERS. Write your answers on the question paper itself, within the blank space provided just after question. All pen-paper calculations needs to be done in the space provided, submit only the filled in question paper. Answer all questions.

You may use your laptop in offline mode. However, any use of *generative AI*, whether online or offline, is strictly prohibited.

Name:
Roll number:

1. $[(2+2)+(2+2+2+2)=10]$

(a) Suppose the final exam scores of 12 students of a subject are 92, 24, 58, 67, 84, 73, 71, 97, 92, 63, 55 and 91.

- i. Calculate average marks of the students and average-student's marks and report those values.
- ii. Find the lower quartile (Q1) and upper quartile (Q3) and interpret those values in the terms of percentage of students who scored more than those quartiles.

(b) A stationery wholesaler in your city is comparing two modes of selling graph paper:

- Mode 1 (Rolls): Each roll contains 50 sheets. He buys 12 rolls and calculates the cost per sheet from each roll.
- Mode 2 (Single Sheets): He also purchases 20 single sheets directly.

From his samples he computes:

- Rolls : mean = Rs. 232.5, standard deviation = Rs. 7.5
 - Single sheets: mean = Rs. 4.70, standard deviation = Rs. 0.20
- i. Convert the mean and standard deviation of the price of the roll to the cost per sheet.
 - ii. Which selling mode shows lower absolute variability in cost per sheet?
 - iii. Which selling mode shows lower relative variability in cost per sheet, i.e. measured by the coefficient of variation?
 - iv. If the wholesaler prefers minimal uncertainty in cost, which selling mode should he opt for, and why?

2. Consider the dataset from the Jelly Bean experiment, where 915 participants guessed the number of jelly beans in a jar. The dataset is available in the course's data repository. [1+1+1+2=5]
- (a) Which kind of average is more appropriate to understand the typical guess of the crowd? Explain the reason and report the number.
 - (b) What measure of dispersion one should use here to measure the variability? Explain why and report that number?
 - (c) Explain a way to reduce the impact of extreme guesses in the analysis.
 - (d) The true value of the number of beans in the jar is 1616. Determine about how many people out of 100 got that close as mean and median, respectively.

3. Consider the heart surgery data given in the course data repository and answer the following questions.[2+1.5+1.5=5]
- (a) For the Child 1991 dataset, remove the first row content named “Bristol” and compute the overall average 30-day survival rate across the other hospitals in the dataset.
 - (b) For same data set (a), find the correlation between % 30 days survival and number of operations.
 - (c) For the Child 2012 dataset, compute the average percentage of 30 days rate survival across the hospitals in the data set.

4.

[(2+2+2)+(2+2)=10]

- (a) Consider the age guessing activity where each students in the class were shown 10 photographs and they were asked to guess age of the person in the photographs. The error (actual-guessed) in guessing the age is available on the data repository.
 - i. Determine which photograph (its index) does not match their age the most? Explain why you say so?
 - ii. Compute the standard deviation and bias for each photograph, and report only the photographs that show the highest standard deviation and the highest bias.
 - iii. Why does the standard deviation not vary much across the photographs, while the bias varies more noticeably?
- (b) Consider the global population data given the data repository and answer the following questions
 - i. Relative to the year 1950, identify the years when the world population doubled and tripled. By plotting the global population in R (or otherwise), make a rough sketch of the world population trend on the answer sheet.
 - ii. Identify the approximate year when Africa became the second most populous continent.